JORGE TORRES

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EDUCATION

The Ohio State University, Columbus OH, USA May 2021
Ph.D. in Physics–Advisor: Prof. Amy Connolly
Master of Science in Physics, July 2017

Universidad de Colima, Colima, Mexico. August 2015
Bachelor of Science in Physics–Advisor: Alfredo Aranda

CURRENT POSITION

University of Utah, Salt Lake City, Utah, USA **July 2025 –**
Postdoctoral Researcher, IceCube Experiment in Carsten Rott's and Dennis Soldin's groups.

PREVIOUS POSITIONS

Yale University, New Haven, Connecticut, USA **July 2021 – July 2025**
Postdoctoral Researcher, Neutrinoless double beta decay (CUORE/CUPID experiments) in Karsten Heeger's/Reina Maruyama's Lab.

The Ohio State University, Columbus, OH, USA **Fall 2015 – June 2021**
PhD Student, Ultra-High Energy Neutrino Astrophysics (Askaryan Radio Array, T-576 experiment)

SELECTED PUBLICATIONS

8. "Reconstruction of cosmic-ray muon events with CUORE"
CUORE Collaboration
[Eur.Phys.J.C 86 (2026) 7, 749].

7. "Performance of a SiPM-based, plastic scintillator muon veto prototype for CUPID"
M. Moore, S. Pagan, E. Pottebaum, J. A. Torres *et. al.*
[2025 JINST 20 P08020] .

6. "Constraints on lepton number violation with the 2 tonne·year CUORE Dataset"
CUORE Collaboration
Science **0**, eadp6474 DOI:10.1126/science.adp6474 .

5. "Snowmass Early Career"
Snowmass Early Career (SEC) organization
[arXiv:2210.12004].

4. "Constraints on the Diffuse Flux of Ultra-High Energy Neutrinos from Four Years of Askaryan Radio Array Data in Two Stations"
P. Allison *et. al.*
Phys. Rev. D 102, 043021 (2020) [arXiv:1912.00987].

3. "Observation of Radar Echoes From High-Energy Particle Cascades"
S. Prohira *et. al.*
Phys Rev Lett. 2020 Mar 6;124(9):091101. [arXiv:1910.12830].

2. "NuRadioMC: Simulating the radio emission of neutrinos from interaction to detector"
C. Glaser *et. al.*
Eur.Phys.J. C80 (2020) no.2, 77. [arXiv:1906.01670].

1. "Suggestion of Coherent Radio Reflections from an Electron-Beam Induced Particle Cascade"
   S.Prohira *et. al.*
   Phys. Rev. D 100, 072003 (2019). [arXiv:1810.09914].

INVITED TALKS

14. Invited talk, University of Utah HEAP Seminar
    Salt Lake City, Utah, USA. 2025/12/05
    Searching for $0\nu\beta\beta$ decay with CUORE and CUPID

13. Invited talk, XIX Mexican Workshop on Particles and Fields, 2025
    Universidad de Guanajuato, Campus León, México. 2025/10/21
    Searching for $0\nu\beta\beta$ decay with CUORE and CUPID

12. Invited talk, HEP Journal Club
    Universidad Católica del Norte, Antofagasta, Chile. 2025/06/19
    Searching for $0\nu\beta\beta$ decay with CUORE and CUPID...and beyond.

11. Invited talk, HEP-ex seminar
    Johns Hopkins University, Baltimore, US. 2025/04/16
    Searching for $0\nu\beta\beta$ decay with CUORE and CUPID...and beyond.

10. Invited talk, Physics and Math Colloquium
    Universidad de Colima, Mexico (remote). 2024/10/12
    Towards the discovery of matter creation with the CUORE and CUPID experiments.

9. Invited talk, HEP seminar
   University of Maryland, College Park, US. 2024/10/06
   CUORE's latest results

8. Invited talk, 2024 Lake Louise Winter Institute
   Banff, Alberta, Canada. 2024/02/22
   Searching for $0\nu\beta\beta$ decay with CUPID

7. Invited talk, 6th Joint Meeting of the APS Division of Nuclear Physics and the Physical Society of Japan
   Big Island, HI. 2023/12/01
   Searching for $0\nu\beta\beta$ decay with CUORE and CUPID

6. Invited talk, HEP Seminar
   Michigan State University, East Lansing, MI. 2022/11/08
   Searching for $0\nu\beta\beta$ decay with CUORE... and beyond

5. Invited talk, Seminario de Altas Energías
   UNAM, Mexico. 2022/05/25
   Searching for $0\nu\beta\beta$ decay with CUORE

4. Invited talk, GAE (Grupo de Altas Energias) Seminar
   CINVESTAV, Mexico. 2022/02/25
   Searching for $0\nu\beta\beta$ decay with CUORE

3. Invited talk, Wright Lab WIDG Seminar
   Wright Lab, Yale Physics Dept, New Haven CT, USA. 2021/10/26
   Tuning into neutrinos on the radio with the ARA experiment

2. Invited plenary talk, XIX Mexican School of Particles and Fields
   Held remotely due to COVID-19 2021/08/10
   Tuning into neutrinos on the radio

1. Invited talk, UMASS Dartmouth Physics Department Colloquium
   Held remotely due to Covid-19 pandemic. 2020/10/15
   Tuning into neutrinos on the radio

CONTRIBUTED TALKS AND POSTERS

15. Poster, XXXII International Conference on Neutrino Physics and Astrophysics.
UC Irvine, USA. 2026/07/21
Results from the first images of the IceCube Upgrade Camera System.

14. Poster, XXXI International Conference on Neutrino Physics and Astrophysics.
Milan, Italy. 2024/06/18
Reconstruction of muon events with CUORE.

13. Contributed talk, APS DNP Fall Meeting, New Orleans LA. 2022/10/28
CUPID: a next-generation $0\nu\beta\beta$ decay experiment

12. Poster, Seattle Snowmass Summer Meeting, Seattle WA. 2022/07/19
CUPID: a next-generation $0\nu\beta\beta$ decay experiment

11. Poster, Neutrino 2022, Seoul, Korea (Virtual). 2022/05/30
Mitigation of cosmic muon backgrounds for CUPID

10. Contributed talk, APS April Meeting, New York, NY. 2022/04/11
Design of a muon-veto system for the CUPID experiment

9. Contributed talk, APS April Meeting, held remotely due to COVID-19 2021/04/19
Reconstruction of UHE neutrinos with the Askaryan Radio Array (ARA) experiment

8. Poster, Cosmic Rays and Neutrinos in the Multi-Messenger Era
Held remotely due to Covid-19 pandemic. 2020/12/07
Recent results from the Askaryan Radio Array (ARA) experiment

7. Contributed talk, 2020 Graduate Student Summer Seminar Series, Columbus OH. 2020/06/30
Ultra-High Energy Neutrinos: Physics, detection, and recent results from the Askaryan Radio Array (ARA) experiment

6. Contributed talk, APS April Meeting, held remotely due to COVID-19 2020/04/19
Recent results from the Askaryan Radio Array (ARA) experiment

5. Contributed talk, Graduate Student Summer Seminar Series, Columbus OH. 2019/07/17
Ultra-High Energy Neutrinos: Physics and Detection

4. Contributed talk, Radio-Workshop, DESY (Zeuthen), Germany. 2019/06/19
Validation of in-ice simulations

3. Contributed talk, APS April Meeting, Denver CO. 2019/04/15
Simulations of radio-based Ultra-High Energy (UHE) in-ice neutrino experiments

2. Contributed talk, Ohio Supercomputer Center Statewide Users Group Conference,
Columbus, OH. 2018/04/05
The role of HPC in the radio-detection of astrophysical neutrinos

1. Contributed talk, Computing in High Energy Astropart. Phys. Research 2016,
Columbus OH. 2016/05/26
The BuckArray: detecting cosmic rays with cellphones

OUTREACH TALKS

11. Science on Tap, by Utah Science Communicators
Salt Lake City UT, USA. 2026/01/31
Neutrinos: the ghost-like particles

10. Pint of Postdoc, by the Yale Postdoctoral Association
New Haven CT, USA. 2023/05/31
Neutrinos: the ghost-like particles

9. Union County Magnet High School
   NJ, USA. 2023/05/05
   Neutrinos: the ghost-like particles

8. Instituto Heisenberg
   Universidad de Colima, Colima, Mexico. 2023/04/01
   Neutrinos: las partículas cuasi-fantasmas

7. Snowmass Summer Study Physics Slam [**winner**]
   University of Washington, Seattle, WA. 2022/07/21
   Neutrinos: the ghost-like particles

6. Girls Advancing in STEM (GAINS) Conference
   Yale University, New Haven, CT. 2022/04/08
   Neutrinos: the ghost-like particles

5. Yale Science in the News talk (series: Hidden things)
   Brookfield Library, Connecticut, USA. 2022/02/24
   ν's from outer space, observing the Universe with neutrinos

4. Yale Science in the News talk (series: Hidden things)
   SSILL, Connecticut, USA. 2022/02/23
   ν's from outer space, observing the Universe with neutrinos

3. Yale Science in the News talk (series: Hidden things)
   New Canaan Library, Connecticut, USA. 2022/02/17
   ν's from outer space, observing the Universe with neutrinos

2. Yale Science in the News talk (series: Hidden things)
   Brooklyn Public Library, New York, USA. 2022/02/15
   ν's from outer space, observing the Universe with neutrinos

1. Talk (high school students), Instituto Heisenberg
   Universidad de Colima, Colima, Mexico . 2019/05/19
   Stories from a physics PhD student in the US

RELEVANT SKILLS

Programming/Software	C++, C, Python, BASH, LaTeX, Git, Geant4, ROOT
Languages	Spanish (Native), English (Full professional proficiency)

AWARDS

- Winner of the Snowmass Summer Study Physics Slam 07/2022
- Selected poster at the Hayes Research Forum 02/2020
  My abstract was selected among two hundred other abstracts to participate in the research forum and present a poster on my work.
- APS Division of Astrophysics Travel Grant to attend the APS April Meeting 04/2019
  My abstract was selected, and I was awarded $600 (USD) to cover travel expenses for the APS April Meeting.
- Ohio SuperComputer Center Statewide Users Group Conference Talk Award 10/2017
  I received this award for getting second place in their 5-minute talk competition.

TEACHING

Teaching Assistant, "Physics 1201:E&M, Optics and Quantum Mechanics", OSU Spring 2018–Summer 2018
Teaching Assistant, "Physics 1250: Mech, Thermo, Waves", OSU Fall 2015–Spring 2017

REVIEWING ACTIVITIES

- Reviewer for Astroparticle Physics

OUTREACH AND SERVICE

Organizer of Salt Lake City's 2026 Pint of Science	May 18-20, 2026
Organizer of New Haven's 2025 Pint of Science	May 20-21, 2025
Panelist at "Science communication for busy scientists" at the 2025 SACNAS New England Chapter Community Gathering	April 12, 2025
Yale Nuclear, Particle and Astrophysics (NPA) seminar coordinator	Aug 2024 –Jan 2025
Panelist at "How to prepare for scientific conferences" workshop at Wright Lab	July 10, 2024
Organizer of New Haven's 2024 Pint of Science	May 13-15, 2024
Volunteer for the 2024 Yale Physics Olympics	March 30, 2024
Member of Postdoctoral Advisory Committee at Yale's Physics Department	Nov, 2023 – Jul, 2025
Panelist at "How to prepare for scientific conferences" workshop at Wright Lab	July 18, 2023
Executive member of the 2023 Yale Physics Olympics	April, 2023
Mentor at the APS DNP Conference Experience for Undergraduates	October, 2022
Postdoc representative for the Yale Physics Club Committee	October, 2022 – October 2023
Yale Science In The News (SITN) Director	September, 2022 – August, 2023
Project lead for Yale Pathways to Science	July 13th, 2022
Speaker at Girls Advancing in STEM (GAINS) Conference	April 8th, 2022
Organizer of "Big Questions in Particle Physics" Snowmass Colloquia	October 2021 – June 2022
Member of the Snowmass Early Career Core Initiatives Leadership	September 2021 – August 2022
Creator and organizer of the *Elusives Journal Club* at Yale Wright Lab	August 2021 – January 2023
Delegate, Council of Graduate Students (CGS), OSU	August 2019–August 2020
Volunteer for "Friends of Ohio State Astronomy and Astrophysics" (FOSAA) event	October 2019
Volunteer for Breakfast of Science Champions, OSU	November 2019
Volunteer Poster Judge, Ohio Supercomputer Center	April 2018–August 2020
Counsel member for the Society for Women in Physics (SWiP), OSU	August 2017–December 2018
Coordinator for ASPIRE Workshop for High School Girls, OSU	July 2017–July 2020

MENTORSHIP

At University of Utah:

Graduate Students:	Seo Won Choi, Shouvik Mondal, Rumman Neshat.
Undergraduate Students:	Marshall Bradley, Ethan Hood, Connor Houston, Nathan Mellish, Porter Pattberg.

At Yale:

Graduate Students:	Ridge Liu, Maya Moore, Samantha Pagan, Iris Ponce, Emily Pottebaum, Ruoxi Wang (student at Johns Hopkins University).
Undergraduate Students:	Iffat Zarif, Aaron Chizhik (now PhD student at UC Berkeley), Din-Ammar Tolj, Andrew Hutchison (student at MIT), Jackie Hua, Andrew Zheng, Reagan Garcia (summer student from CalPoly).

At Ohio State:

Graduate Students:	Dennis Calderon-Madera, Julie Rolla, Justin Flaherty, Dylan Frikken
Undergraduate Students:	Ian Best, Hannah Hassan, Alex Machtay, Alex Patton

SOFTWARE PROJECTS

- Lobster Plot: Code to make the so-called "Lobster Plot", developed from code that previous Yale grad. student Jeremy Cushman wrote. Website: `https://toej93.github.io/LobsterPlot/`.

IN THE MEDIA

- Wright Lab members host 2025 Pint of Science Festival in New Haven.
- Yale Physics postdocs host 2024 Pint of Science Festival in New Haven
- Building on luck, Symmetry Magazine
- Statewide Users Group spring conference showcases variety of research, OSC resources

Full publication list (without proceedings)

[1] Maya Moore et al. "Performance of a SiPM-based, plastic scintillator muon veto prototype for CUPID". In: (May 2025). arXiv: 2505.06129 [physics.ins-det].

[2] D. Q. Adams et al. "Data-driven background model for the CUORE experiment". In: *Phys. Rev. D* 110.5 (2024), p. 052003. DOI: 10.1103/PhysRevD.110.052003. arXiv: 2405.17937 [nucl-ex].

[3] D. Q. Adams et al. "Search for Fractionally Charged Particles with CUORE". In: *Phys. Rev. Lett.* 133 (24 Dec. 2024), p. 241801. DOI: 10.1103/PhysRevLett.133.241801. URL: https://link.aps.org/doi/10.1103/PhysRevLett.133.241801.

[4] CUORE Collaboration et al. *With or without ν? Hunting for the seed of the matter-antimatter asymmetry*. 2024. arXiv: 2404.04453 [nucl-ex]. URL: https://arxiv.org/abs/2404.04453.

[5] B. Acharya et al. "Fundamental Symmetries, Neutrons, and Neutrinos (FSNN): Whitepaper for the 2023 NSAC Long Range Plan". In: (Apr. 2023). arXiv: 2304.03451 [nucl-ex].

[6] J. A. Aguilar et al. "Triboelectric backgrounds to radio-based polar ultra-high energy neutrino (UHEN) experiments". In: *Astropart. Phys.* 145 (2023), p. 102790. DOI: 10.1016/j.astropartphys.2022.102790.

[7] K. Alfonso et al. "A first test of CUPID prototypal light detectors with NTD-Ge sensors in a pulse-tube cryostat". In: *JINST* 18.06 (2023), P06033. DOI: 10.1088/1748-0221/18/06/P06033. arXiv: 2304.04674 [physics.ins-det].

[8] K. Alfonso et al. "Twelve-crystal prototype of Li_2MoO_4 scintillating bolometers for CUPID and CROSS experiments". In: *JINST* 18.06 (2023), P06018. DOI: 10.1088/1748-0221/18/06/P06018. arXiv: 2304.04611 [physics.ins-det].

[9] J. A. Aguilar et al. "In situ, broadband measurement of the radio frequency attenuation length at Summit Station, Greenland". In: (Jan. 2022). DOI: 10.1017/jog.2022.40. arXiv: 2201.07846 [astro-ph.IM].

[10] J. A. Aguilar et al. "Reconstructing the neutrino energy for in-ice radio detectors: A study for the Radio Neutrino Observatory Greenland (RNO-G)". In: *Eur. Phys. J. C* 82.2 (2022), p. 147. DOI: 10.1140/epjc/s10052-022-10034-4. arXiv: 2107.02604 [astro-ph.HE].

[11] K. Alfonso et al. "Optimization of the first CUPID detector module". In: *Eur. Phys. J. C* 82.9 (2022), p. 810. DOI: 10.1140/epjc/s10052-022-10720-3. arXiv: 2202.06279 [physics.ins-det].

[12] P. Allison et al. "Low-threshold ultrahigh-energy neutrino search with the Askaryan Radio Array". In: *Phys. Rev. D* 105.12 (2022), p. 122006. DOI: 10.1103/PhysRevD.105.122006. arXiv: 2202.07080 [astro-ph.HE].

[13] Joshua Barrow et al. "Snowmass Early Career: The Key Initiatives Organization". In: *Snowmass 2021*. July 2022. arXiv: 2207.07508 [physics.soc-ph].

[14] J. A. Aguilar et al. "Design and Sensitivity of the Radio Neutrino Observatory in Greenland (RNO-G)". In: *JINST* 16.03 (2021). [Erratum: JINST 18, E03001 (2023)], P03025. DOI: 10.1088/1748-0221/16/03/P03025. arXiv: 2010.12279 [astro-ph.IM].

[15] J. A. Aguilar et al. "Triboelectric Backgrounds to radio-based UHE Neutrino Exeperiments". In: (Mar. 2021). arXiv: 2103.06079 [astro-ph.IM].

[16] S. Prohira et al. "Modeling in-ice radio propagation with parabolic equation methods". In: *Phys. Rev. D* 103.10 (2021), p. 103007. DOI: 10.1103/PhysRevD.103.103007. arXiv: 2011.05997 [astro-ph.IM].

[17] S. Prohira et al. "The Radar Echo Telescope for Cosmic Rays: Pathfinder experiment for a next-generation neutrino observatory". In: *Phys. Rev. D* 104.10 (2021), p. 102006. DOI: 10.1103/PhysRevD.104.102006. arXiv: 2104.00459 [astro-ph.IM].

[18] M.G. Aartsen et al. "IceCube-Gen2: The Window to the Extreme Universe". In: (Aug. 2020). arXiv: 2008.04323 [astro-ph.HE].

[19] P. Allison and others [co-author]. "Constraints on the diffuse flux of ultrahigh energy neutrinos from four years of Askaryan Radio Array data in two stations". In: *Phys. Rev. D* 102.4 (2020), p. 043021. DOI: 10.1103/PhysRevD.102.043021. arXiv: 1912.00987 [astro-ph.HE].

[20] Christian Glaser et al. "NuRadioMC: Simulating the radio emission of neutrinos from interaction to detector". In: *Eur. Phys. J. C* 80.2 (2020), p. 77. DOI: 10.1140/epjc/s10052-020-7612-8. arXiv: 1906.01670 [astro-ph.IM].

[21] S. Prohira et al. "Observation of Radar Echoes From High-Energy Particle Cascades". In: *Phys. Rev. Lett.* 124.9 (2020), p. 091101. DOI: 10.1103/PhysRevLett.124.091101. arXiv: 1910.12830 [astro-ph.HE].

[22] M. G. Aartsen et al. "Neutrino astronomy with the next generation IceCube Neutrino Observatory". In: (Nov. 2019). arXiv: 1911.02561 [astro-ph.HE].

[23] J.A. Aguilar et al. "The Next-Generation Radio Neutrino Observatory – Multi-Messenger Neutrino Astrophysics at Extreme Energies". In: (July 2019). arXiv: 1907.12526 [astro-ph.HE].

[24] P. Allison et al. "Long-baseline horizontal radio-frequency transmission through polar ice". In: *arXiv e-prints*, arXiv:1908.10689 (Aug. 2019), arXiv:1908.10689. arXiv: 1908.10689 [astro-ph.IM].

[25] S. Prohira et al. "Suggestion of Coherent Radio Reflections from an Electron-Beam Induced Particle Cascade". In: *Phys. Rev. D* 100.7 (2019), p. 072003. DOI: 10.1103/PhysRevD.100.072003. arXiv: 1810.09914 [hep-ex].